No zeros, for Riemann hypothesis on critical strip

Problem

The Riemann hypothesis is concerned with the locations of non-trivial zeros, and states that:

The real part of every non-trivial zero of the Riemann zeta function is $\frac{1}{2}$.

Abstract

This paper, tries to address the riemann hypothesis, using algebra.

Begin of proof

We have the zeta function, defined in piece wise format. It is the following

$$\zeta(s) = \begin{cases} \sum_{n=1}^{\infty} \frac{1}{n^s}, & \operatorname{Re}(s) > 1, \\ \frac{1}{1-2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s}, & 0 < \operatorname{Re}(s) < 1, \\ 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \sum_{n=1}^{\infty} \frac{1}{n^{1-s}}, & \operatorname{Re}(s) < 0. \end{cases} \tag{1}$$

Since we are only interessed, in critical strip we choose the second function.

So from (1) , we get (2)

For $0 < \operatorname{Re}(s) < 1$

$$\zeta(s) = \frac{1}{1-2^{1-s}} \sum_{n=1}^{\infty} \begin{cases} \frac{1}{n^s}, & \text{if n is odd,} \\ -\frac{1}{n^s}, & \text{if n is even.} \end{cases} \tag{2}$$

Assuming $\zeta(s)$= 0. Dividing both sides by $\frac{1}{1-2^{1-s}}$ we get

$$0 = \sum_{n=1}^{\infty} \begin{cases} \frac{1}{n^s}, & \text{if n is odd,} \\ -\frac{1}{n^s}, & \text{if n is even.} \end{cases}$$

So, we have

$$0 = \frac{1}{1^s} - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \frac{1}{5^s} \cdots . \tag{3}$$

Rearranje (3) equation

$$0 = 1 - (\frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} \cdots) + (\frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} \cdots)$$

Subtract -1 by both sides

$$-1 = -(\frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} \cdots) + (\frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} \cdots)$$

Change second summand, to left, and change sign

$$-1 - \frac{1}{3^s} - \frac{1}{5^s} - \frac{1}{7^s} - \cdots = -\frac{1}{2^s} - \frac{1}{4^s} - \frac{1}{6^s} - \cdots$$

Multiply both sides by -1

$$1 + \frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \cdots = \frac{1}{2^s} + \frac{1}{4^s} + \frac{1}{6^s} + \cdots$$

Subtract both sides, by 1

$$\frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \cdots = \frac{1}{2^s} - 1 + \frac{1}{4^s} + \frac{1}{6^s} \cdots$$

$\frac{1}{2^s}-1$ is negative. So substitude right side by that.

$$\frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \cdots = \text{negative value} + \frac{1}{4^s} + \frac{1}{6^s} + \frac{1}{8^s} + \cdots$$

putting all terms, on right side (except the negative value), to the left

$$\frac{1}{3^s} - \frac{1}{4^s} + \frac{1}{5^s} - \frac{1}{6^s} + \frac{1}{7^s} - \frac{1}{8^s} - \cdots = \text{negative value} \tag{4}$$

So, on left side, we have a positive value, because , considering two elements,

$\frac{1}{x} - \frac{1}{x+1}$, is always positive, since minuend is bigger that subtrahend (and numerator is the same)

Example

$\frac{1}{3^s} - \frac{1}{4^s}$, will always give positive.

So, on (4), left side, is positive. But we have negative value on right side. Doesn't hold.

Conclusion

There is no zeros, on critical strip of zeta function.

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Bibliography

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